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Round bale pre-completion yield calculation

Abstract

A round baler includes a bale size sensor operable to detect a radial size of a bale during formation thereof, a bale weight sensor operable to detect a weight of a completed bale, and a location sensor operable to determine a respective field origin location of the crop material gathered for each respective one of a plurality of radial layers of the bale. A controller calculates a density of a completed bale with data from the bale weight sensor and the bale size sensor. The controller determines a volume of a respective one of the radial layers during formation of a current bale, subsequent to the completed bale, with data from the bale size sensor. The controller may then estimate a density of the respective one of the radial layers based on the volume of the respective one of the radial layers, and the density of the previously completed bale.

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Background/Summary

TECHNICAL FIELD

(1) The disclosure generally relates to a round baler system, and a method of generating a yield map of crop material formed into a round bale with the round baler system.

BACKGROUND

(2) Agricultural balers gather crop material from a field, and compress, the crop material into a bale. A round baler implement shapes the crop material into a cylindrical shape, typically referred to as a round bale. The round baler implement feeds a continuous supply of crop material into a baling chamber to form a round bale. Because of this, the quantity or volume of the crop material required to form a round bale may be gathered from a large area spread throughout the field.

(3) Various crop and harvest management operations may benefit from an accurate yield map showing crop yield across the field. The yield may be considered mass of the crop material per unit area of the field, i.e., a density of the harvested crop material. The density of the crop material may be calculated by multiplying a known or determined volume by a known or determined weight. For

a round bale, the density of the crop material is generally determined by sensing a size and weight of a completed round bale. These values may then be multiplied to calculate the density of the completed round bale. As such, the yield map is limited to the density of the completed round bale as a whole spread across the entire region of the field from which the crop material was gathered for that specific round bale.

(4) Yield maps based on this average bale density described above are therefore generalized and lack the desired level of detail or specificity. Additionally, the yield maps described above are latent because they require completion of a specific bale to calculate the yield from that specific bale. As such, the generation of the yield map is post processing of the completed round bale, which may reduce the value of the yield map for certain harvest operations.

SUMMARY

(5) A round baler system is provided. The round baler system includes a pick-up operable to gather and move crop material while traversing a field. A baler system having a baling chamber configured to receive a flow of crop material from the pick-up and roll the incoming flow of crop material in a spiral fashion into a bale having a cylindrical shape. The bale includes a plurality of radial layers. A bale size sensor is operable to detect data related to a radial size of the bale during formation thereof. A bale weight sensor is operable to detect data related to a weight of a completed bale. A location sensor is operable to determine a respective field origin location of the crop material gathered by the pick-up for each respective one of the plurality of radial layers of the bale. A controller is disposed in communication with the bale size sensor, the bale weight sensor, and the location sensor. The controller includes a processor and a memory having a yield mapping algorithm stored thereon. The processor is operable to execute the yield mapping algorithm to calculate a density of a completed bale with data received from the bale weight sensor and the bale size sensor. The controller is operable to determine a volume of a respective one of the plurality of radial layers during formation of a current bale subsequent to the completed bale with data received from the bale size sensor. The controller may then estimate a density of the respective one of the plurality of radial layers during formation of the current bale based on the volume of the respective one of the plurality of radial layers, and the density of the completed bale.

(6) In one aspect of the disclosure, the baler system may further include a moisture sensor that is operable to detect data related to a moisture content of the crop material for each respective one of the plurality of radial layers. The processor may be operable to execute the yield mapping algorithm to determine a moisture content of the crop material of the respective one of the plurality of radial layers during formation of the current bale with data received from the moisture sensor, and estimate the density of the respective one of the plurality of radial layers during formation of the current bale based on the volume of the respective one of the plurality of radial layers, the moisture content of the respective one of the plurality of radial layers, and the density of the completed bale.

(7) In one aspect of the disclosure, the processor is operable to execute the yield mapping algorithm to estimate the density of the respective one of the plurality of radial layers during formation of the current bale based on the density of at least two previously completed bales. For example, the processor may use a running average or a weighted average of the prior 2, 3, . . . , n, fully completed bales to estimate the density of the radial layer of the current bale during formation thereof.

(8) In one aspect of the disclosure, the processor is operable to execute the yield mapping algorithm to estimate a density of each of the plurality of radial layers of the current bale during formation of the current bale. Accordingly, the controller may estimate the yield of each respective radial layer based on the moisture content of that respective layer, the volume of that respective layer, and the density of one or more previously completed bales. By doing so, the detail and granularity of the yield data is greatly increased.

(9) In one aspect of the disclosure, the processor is operable to execute the yield mapping algorithm

to determine a field origin location of the respective one of the plurality of radial layers during formation of the current bale. The processor may be operable to execute the yield mapping algorithm to generate a yield map of the field based on the estimated density and the field origin location of the respective one of the plurality of radial layers during formation of the current bale. As such, the yield map is generated based on the yield of each individual radial layer of the bale and the associated portion of the field from which crop material was gathered for that specific radial layer. By so doing, the detail of the yield map is greatly increased from previous yield map data based on the average yield of the entire completed bale.

(10) In one aspect of the disclosure, the processor may be operable to execute the yield mapping algorithm to control a function of the baler system based on the yield map of the field. Because the yield map provides highly detailed data during formation of the bale, instead of post bale processing, the yield map may be used to predict harvest ahead of the baler system, and thereby control the operation of the baler system in real time. For example, the function of the baler system may include path planning for the baling operation, weaving control of the baler system to evenly distribute crop material across a width of the baling chamber, a speed of movement across the field, a secondary crop processing function, a forming belt tension, a bale drop location, etc.

(11) In one aspect of the disclosure, a method of generating a yield map of crop material formed into a round bale is provided. The method includes completely forming a first bale in a baling chamber of a baler system. A density of the first bale may then be calculated after the first bale is completely formed. A second bale may then be formed in the baling chamber of the baler system. The second bale is formed via a plurality of radial layers radially stacked one on top of another. A volume of a respective one of the plurality of radial layers of the second bale may then be determined during formation of the second bale, and subsequent to the completed first bale, with data received from a bale size sensor. A moisture content of the crop material of the respective one of the plurality of radial layers of the second bale may be determined during formation of the second bale with data received from a moisture sensor. A density of the respective one of the plurality of radial layers of the second bale may then be estimated during formation of the second bale based on the volume and the moisture content of the respective one of the plurality of radial layers of the second bale, and the density of the completed first bale. A field origin location of the respective one of the plurality of radial layers of the second bale is determined during formation of the second bale. The yield map of the field may then be generated by a controller based on the estimated density and the field origin location of the respective one of the plurality of radial layers of the second bale during formation of the second bale.

(12) In one aspect of the controller the method described herein may further include controlling a function of the baler system based on the yield map of the field. Because the yield map provides highly detailed data during formation of the bale, instead of post bale processing, the yield map may be used to predict harvest ahead of the baler system, and thereby control the operation of the baler system in real time. For example, the function of the baler system may include path planning for the baling operation, weaving control of the baler system to evenly distribute crop material across a width of the baling chamber, a speed of movement across the field, a secondary crop processing function, a forming belt tension, a bale drop location, etc.

(13) The above features and advantages and other features and advantages of the present teachings are readily apparent from the following detailed description of the best modes for carrying out the teachings when taken in connection with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic side view of a round baler system.

- (2) FIG. 2 is a schematic perspective view of a round bale
(3) FIG. 3 is a schematic end view of the round bale.
(4) FIG. 4 is a schematic plan view of a yield map representing a field.

DETAILED DESCRIPTION

- (5) Those having ordinary skill in the art will recognize that terms such as “above,” “below,” “upward,” “downward,” “top,” “bottom,” etc., are used descriptively for the figures, and do not represent limitations on the scope of the disclosure, as defined by the appended claims. Furthermore, the teachings may be described herein in terms of functional and/or logical block components and/or various processing steps. It should be realized that such block components may be comprised of any number of hardware, software, and/or firmware components configured to perform the specified functions.
- (6) Terms of degree, such as “generally”, “substantially” or “approximately” are understood by those of ordinary skill to refer to reasonable ranges outside of a given value or orientation, for example, general tolerances or positional relationships associated with manufacturing, assembly, and use of the described embodiments.
- (7) As used herein, “e.g.” is utilized to non-exhaustively list examples, and carries the same meaning as alternative illustrative phrases such as “including,” “including, but not limited to,” and “including without limitation.” As used herein, unless otherwise limited or modified, lists with elements that are separated by conjunctive terms (e.g., “and”) and that are also preceded by the phrase “one or more of,” “at least one of,” “at least,” or a like phrase, indicate configurations or arrangements that potentially include individual elements of the list, or any combination thereof. For example, “at least one of A, B, and C” and “one or more of A, B, and C” each indicate the possibility of only A, only B, only C, or any combination of two or more of A, B, and C (A and B; A and C; B and C; or A, B, and C). As used herein, the singular forms “a”, “an” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise. Further, “comprises,” “includes,” and like phrases are intended to specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof.
- (8) Referring to FIG. 1, a baler system is generally shown at **20**. The baler system **20** may alternatively be referred to herein as a round baler **20**. The round baler **20** includes a frame **22**. One or more ground engaging elements **24**, such as but not limited to one or more wheels and/or tracks, are attached to and rotatably supported by the frame **22**. A tongue **26** may be coupled to the frame **22** at a forward end of the frame **22**. A hitch arrangement **28** may be included with the tongue **26**. The hitch arrangement **28** may be used to attach the round baler **20** to a traction unit **30**, such as but not limited to an agricultural tractor. In other embodiments, the round baler **20** may be self-propelled, in which case the traction unit **30** and the round baler **20** are configured as a single, self-propelled vehicle.
- (9) The round baler **20** includes a housing **32** forming a baling chamber **34**. The housing **32** is attached to and supported by the frame **22**. The housing **32** may include one or more walls or panels that at least partially enclose and/or define the baling chamber **34**. The round baler **20** further includes a gate **36**. The gate **36** is attached to and rotatably supported by the housing **32**. The gate **36** is positioned adjacent a rearward end of the frame **22** and is pivotably moveable about a gate axis **38**. The gate axis **38** is generally horizontal and perpendicular to a central longitudinal axis **40** of the frame **22**. The gate **36** is moveable between a closed position for forming a bale **100** within the baling chamber **34**, and an open position for discharging the bale **100** from the baling chamber **34**.
- (10) The round baler **20** includes a pick-up **42** positioned proximate the forward end of the frame **22**. The pickup gathers crop material from a ground surface **44** of a field **46** as the round baler **20** traverses the field **46**. The pickup directs the gathered crop material toward and into an inlet **52** of the baling chamber **34**. The pickup may include, but is not limited to tines, forks, augers,

conveyors, baffles, etc., for gathering and moving the crop material. The round baler **20** may be equipped with a pre-cutter, disposed between the pickup and the inlet **52** of the baling chamber **34**. As such, the pre-cutter is disposed downstream of the pickup and upstream of the inlet **52** relative to a direction of travel of the crop material. The pre-cutter cuts or chops the crop material into smaller pieces.

(11) The round baler **20** may be configured as a variable chamber baler, or as a fixed chamber baler. The round baler **20** shown in FIG. **1** and described herein is depicted and described as a variable chamber baler. As is understood by those skilled in the art, the variable chamber baler includes a plurality of longitudinally extending side-by-side forming belts **48** that are supported by a plurality of rollers **50**. The bale **100** is formed by the forming belts **48** and one or more side walls of the housing **32**.

(12) The crop material is directed through the inlet **52** and into the baling chamber **34**, whereby the forming belts **48** roll the crop material in a spiral fashion into the bale **100** having a cylindrical shape. The belts apply a constant pressure to the crop material as the crop material is formed into the bale **100**. A belt tensioner **54** continuously moves the forming belts **48** radially outward relative to a center of the cylindrical bale **100** as the diameter **58** of the bale **100** increases. The belt tensioner **54** maintains the appropriate tension in the belts to obtain the desired density of the crop material.

(13) Referring to FIGS. **2** and **3**, the cylindrical shape of the formed bale **100** includes a center axis **56** of the bale **100**. Circular end faces **102** of the bale **100** are positioned at axial ends of the bale **100** along the center axis **56**. The circular end faces **102** of the bale **100** include and/or define a diameter **58** of the bale **100**. The bale **100** includes an exterior circumferential surface **104** extending between the circular end faces of the bale **100** a length **60**. The length **60** of the bale **100** is generally fixed and/or defined by a width of the baling chamber **34**. However, the diameter **58** of the bale **100** may be controlled by the baler system **20**.

(14) The bale **100** may be sectioned and/or partitioned into a plurality of radial layers **62**. Each radial layer **62** includes a radial thickness **64** extending transverse to the center axis **56** of the bale **100**, and extends the length **60** of the bale **100** between the circular end faces **102** of the bale **100**. As such, a first radial layer **62** may be defined at the center of the bale **100** by a solid cylindrical shape, whereas each progressive outer radial layer is positioned radially outward from the first radial layer **62**, and is defined by a hollow cylindrical shape. For example, a second radial layer **62** is shown at **62B**, a third radial layer **62** is shown at **62C**, a fourth radial layer **62** is shown at **62D**, etc. The bale **100** may be partitioned into any number of radial layers **62**. A greater number of the radial layers **62** generally corresponds to a higher level of resolution and increased accuracy of a yield map **98**, described in greater detail below. The radial thickness **64** of each respective radial layer **62** may be defined as a constant, or may be varied, such that each of the respective radial layers **62** has a different radial thickness **64**. In other implementations, the radial layers **62** may be grouped into sets, with each respective radial layer **62** within a given set of layers having a similar radial thickness **64**, and different sets of radial layers **62** having a substantially different radial thickness **64**. In one example implementation, the bale **100** may be partitioned into between ten and twenty radial layers **62**.

(15) In another implementation the bale **100** may be portioned into the plurality of radial layers **62** based on the formation of the bale **100**. As described above, the bale **100** is formed in a spiral fashion by rotating the bale **100** within the baling chamber **34**. As such, each respective radial layer **62** may be defined based on a single rotation of the bale **100** within the baling chamber **34**. In other words, each rotation of the bale **100** in the baling chamber **34** forms a successive radial layer **62** of the bale **100**. The radial thickness **64** of the radial layers **62** defined in such a manner may be variable, based on the amount of crop material used in the formation of that respective radial layer **62** and the density of the crop material in that respective radial layer **62**.

(16) Referring to FIG. **1**, the round baler **20** includes a wrap system **66**. The wrap system **66** is

operable to wrap the bale **100** with a wrap material **68** inside the baling chamber **34**. Once the bale **100** is formed to a desired size, the wrap system **66** feeds the wrap material **68** into the baling chamber **34** to wrap the bale **100** and thereby secure the crop material in a tight package and maintain the desired shape of the bale **100**. The wrap material **68** may include, but is not limited to, a twine, a net mesh, or a solid plastic wrap. Movement of the gate **36** into the open position simultaneously moves the belts clear of the formed bale **100**, and allows the formed and wrapped bale **100** to be discharged through the rear of the baling chamber **34**.

(17) The baler system **20** includes a bale size sensor **70** that is operable to detect data related to a radial size of the bale **100** during formation thereof. In one implementation, the bale size sensor **70** is operable to detect the radial size of each radial layer **62** of the bale **100** during formation of the bale **100**. For example, if the radial layers **62** are defined to include the thickness of the layer formed during a single rotation of the bale **100** in the baling chamber **34**, then the bale size sensor **70** may detect data related to the radial size of the bale **100** at the beginning and end of each rotation of the bale **100** in the baling chamber **34**. In another implementation, the bale size sensor **70** may include location and time data related to the radial size of the bale **100** during formation, such that the location and time of each respective radial layer **62** of the bale **100** may be determined by a controller **90**, described in greater detail below. The baler size sensor may include a device that is capable of detecting or sensing data related to the radial size of the bale **100**. For example, the bale size sensor **70** may include, but is not limited to, and optical sensor positioned to capture an image of the bale **100** from which the radial size of the bale **100** may be determined, or position sensor, e.g., a potentiometer, that is positioned and arranged to detect a position of a component of the baler system **20** related to the radial size of the bale **100**, such as but not limited to an belt tensioning arm of the baler system **20**. It should be appreciated that the bale size sensor **70** may include some other device that senses some other type of data not described herein that may be used to determine the radial size of the bale **100**.

(18) The baler system **20** further includes a bale weight sensor **72**. The bale weight sensor **72** is operable to detect data related to a weight of the bale **100**. In one implementation, the bale weight sensor **72** may detect the weight of at least the completed bale **100**. However, in other implementations, the baler weight sensor may detect the weight of the bale **100** from beginning through completion, i.e., detect the changing weight of the bale **100** during formation and at completion. The bale weight sensor **72** may include a device that is capable of detecting data related to the weight of the bale **100**. For example, the bale weight sensor **72** may include a force sensor positioned and arranged to detect a weight of the while in the baling chamber **34**, such as detecting a force and/or strain on an axle or component thereof. In other implementations, the bale weight sensor **72** may include a force sensor operable to detect the force and/or strain of a component used to hold or lift the bale **100**. For example, the bale weight sensor **72** may include a force sensor positioned and operable to detect a force exerted by a hydraulic cylinder associated with the lift gate **36** used to lift the completed bale **100** out of the baling chamber **34** post completion. It should be appreciated that the bale weight sensor **72** may include some other device that senses some other type of data not described herein that may be used to determine the weight of the bale **100**.

(19) The baler system **20** may further include a moisture sensor **74** that is operable to detect data related to a moisture content of the crop material for each respective one of the plurality of radial layers **62**. The moisture sensor **74** may include for example, but is not limited to, a Near Infra-Red light spectrum optical sensor capable of detecting images in the NIR light spectrum, which may be used to determine the moisture content of the crop material. The moisture sensor **74** may be positioned to detect the moisture level of the crop material either in the baling chamber **34**, or outside the baling chamber **34** in the flow of crop material entering the baling chamber **34**, such as proximate the inlet **52** of the baling chamber **34**. The data from the moisture sensor **74** may associated with the specific crop material used to form a respective radial layer **62** of the bale **100**.

For example, with the beginning of each rotation of the bale **100** within the baling chamber **34**, the moisture sensor **74** data may be associated with the respective radial layer **62** of the bale **100** being formed. Upon the completion of a respective radial layer **62** of the bale **100**, the data from the moisture sensor **74** may be associated the next radial layer **62** of the bale **100**.

(20) As shown in FIG. 1, the baler system **20** further includes a location sensor **76**. Referring to FIG. 4, the location sensor **76** is operable to determine a respective field origin location **78** of the crop material gathered by the pick-up **42** for each respective one of the plurality of radial layers **62**. The field origin location **78** of the crop material for each radial layer **62** may include a coordinate or other positional identification of the region of the field **46** from which the crop material used to form that respective radial layer **62** of the bale **100** was gathered. The field origin location **78** may include, but is not limited to, an boundary **80** of a region outlining the area of the field **46** from which the crop material of that respective radial layer **62** of the bale **100** was gathered, a centroid location **82** of a region of standardized size representing the region outlining the area of the field **46** from which the crop material of that respective radial layer **62** of the bale **100** was gathered, or a beginning location **84** and an ending location **86** of a track **88** having a defined width outlining or representing the area of the field **46** from which the crop material of that respective radial layer **62** of the bale **100** was gathered. The location sensor **76** may include, but is not limited to, a device capable of sensing data that may be used to determine a geographic location within the field **46**. For example, the location sensor **76** may include, but is not limited to, a Global Positioning System sensor that is operable to receive satellite signals and triangulate a position location therefrom as is understood in the art.

(21) As noted above, the baler system **20** includes a controller **90**. The controller **90** is disposed in communication with the bale size sensor **70**, the bale weight sensor **72**, the moisture sensor **74**, and the location sensor **76**. While the controller **90** is generally described herein as a singular device, it should be appreciated that the controller **90** may include multiple devices linked together to share and/or communicate information therebetween. Furthermore, it should be appreciated that the controller **90** may be located on the round baler **20** implement or located remotely from the round baler **20** implement, such as on an associated traction unit **30**, e.g., tractor.

(22) The controller **90** may alternatively be referred to as a computing device, a computer, a control unit, a control module, a module, etc. The controller **90** includes a processor **92**, a memory **94**, and all software, hardware, algorithms, connections, sensors, etc., necessary to manage and control the operation of the baler system **20**. As such, a method may be embodied as a program or algorithm operable on the controller **90**. It should be appreciated that the controller **90** may include any device capable of analyzing data from various sensors, comparing data, making decisions, and executing the required tasks.

(23) As used herein, “controller **90**” is intended to be used consistent with how the term is used by a person of skill in the art, and refers to a computing component with processing, memory, and communication capabilities, which is utilized to execute instructions (i.e., stored on the memory or received via the communication capabilities) to control or communicate with one or more other components. In certain embodiments, the controller **90** may be configured to receive input signals in various formats (e.g., hydraulic signals, voltage signals, current signals, CAN messages, optical signals, radio signals), and to output command or communication signals in various formats (e.g., hydraulic signals, voltage signals, current signals, CAN messages, optical signals, radio signals).

(24) The controller **90** may be in communication with other components on the baler system **20**, such as hydraulic components, electrical components, and operator inputs within an operator station of an associated work vehicle. The controller **90** may be electrically connected to these other components by a wiring harness such that messages, commands, and electrical power may be transmitted between the controller **90** and the other components. Although the controller **90** is referenced in the singular, in alternative implementations the configuration and functionality described herein can be split across multiple devices using techniques known to a person of

ordinary skill in the art.

(25) The controller **90** may be embodied as one or multiple digital computers or host machines each having one or more processors **92**, read only memory (ROM), random access memory (RAM), electrically-programmable read only memory (EPROM), optical drives, magnetic drives, etc., a high-speed clock, analog-to-digital (A/D) circuitry, digital-to-analog (D/A) circuitry, and any required input/output (I/O) circuitry, I/O devices, and communication interfaces, as well as signal conditioning and buffer electronics.

(26) The computer-readable memory **94** may include any non-transitory/tangible medium which participates in providing data or computer-readable instructions. The memory **94** may be non-volatile or volatile. Non-volatile media may include, for example, optical or magnetic disks and other persistent memory. Example volatile media may include dynamic random access memory (DRAM), which may constitute a main memory. Other examples of embodiments for memory **94** include a floppy, flexible disk, or hard disk, magnetic tape or other magnetic medium, a CD-ROM, DVD, and/or any other optical medium, as well as other possible memory devices such as flash memory.

(27) The controller **90** includes the tangible, non-transitory memory **94** on which are recorded computer-executable instructions, including a yield mapping algorithm **96**. The processor **92** of the controller **90** is configured for executing the yield mapping algorithm **96**. The yield mapping algorithm **96** implements a method of generating a yield map **98** and controlling the baler system **20**, described in detail below.

(28) Referring to FIG. **4**, the process begins by completely forming a first bale **106** in the baling chamber **34** of the baler system **20**. The controller **90** may then calculate a density of the completed first bale **106** with data received from the bale weight sensor **72** and the bale size sensor **70**. The bale size sensor **70** may communicate a data signal to the controller **90** including information related to the size of the first bale **106**. In one implementation, the length **60** of the first bale **106** may be presumed based on the width of the baling chamber **34**. The data from the bale size sensor **70** may relate, for example, to the diameter **58** of the first bale **106**. The controller **90** may then calculate a total volume of the first bale **106** from the data related to the diameter **58** of the first bale **106** and the known length **60** of the first bale **106**. The bale weight sensor **72** may communicate a data signal to the controller **90** including information related to the weight of the first bale **106**. The controller **90** may then calculate the weight of the completed first bale **106**. Having determined the volume and weight of the first bale **106**, the controller **90** may then calculate the density of the completed first bale **106** by, for example, dividing the weight by the volume.

(29) Once the first bale **106** has been completely formed, the first bale **106** may be ejected from the baling chamber **34**, after which a second bale **100** may be formed in the baling chamber **34** of the baler system **20**. As described above, the second bale **100** may be segmented into the plurality of radial layers **62**. In one implementation, each of the radial layers **62** may be defined based on a pre-set radial thickness **64**. In another implementation, each of the radial layers **62** may be defined based on a set number of rotations of the bale **100** in the baling chamber **34**, e.g., a single rotation, two rotations, etc., in which case the radial thickness **64** of each layer will vary and is dependent upon the amount or volume of crop material supplied to the baling chamber **34** while forming a particular radial layer **62**.

(30) The process described below may be used for each of the plurality of radial layers **62** in succession during formation of the second bale **100** in the forming chamber. The description below describes the process for a single one of the plurality of radial layers **62**. However, it should be appreciated that the process is repeated for each of the plurality of radial layers **62** of the second bale **100**.

(31) During formation of a current bale **100**, e.g., the second bale **100** which is formed subsequent to the completed first bale **106**, each of the plurality of radial layers **62** is formed in succession.

Upon the completion of a respective one of the radial layers **62** of the second bale **100**, the controller **90** determines a volume of that respective one of the radial layers **62** with data received from the bale size sensor **70**. The data received from the bale size sensor **70** may include, but is not limited to, data related to a radial or diametric size of the second bale **100** at the beginning of the respective radial layer **62** and a radial or diametric size of the second bale **100** at the end or completion of the respective radial layer **62**. Knowing the diameter **58** of the second bale **100** at the beginning and end of the respective radial layer **62**, and knowing the length **60** of the bale **100** (which is generally a standard dimension based on the width of the baling chamber **34**), the controller **90** may calculate the volume of the respective radial layer **62**. It should be appreciated that the data from the bale size sensor **70** may differ from the example implementation described herein, and the process in which the controller **90** calculates or determines the volume of the respective radial layer **62** may differ from the example implementation described herein based on the type of data from the bale size sensor **70**.

(32) The controller **90** may further determine a moisture content of the crop material of the respective one of the radial layers **62** during formation of the current, e.g., the second bale **100**. The controller **90** may determine the moisture content from data received from the moisture sensor **74**. For example, the controller **90** may continuously sense the moisture content of the crop material entering the baling chamber **34**. Based on a defined start time and end time of the formation of the respective radial layer **62**, the controller **90** may associate the moisture content of the crop material measured during these time periods with the respective radial layer **62**, and calculate an average moisture content from the measured values associated with the respective layers. It should be appreciated that the data from the moisture sensor **74** may differ from the example implementation described herein, and the process in which the controller **90** calculates or determines the moisture content of the respective radial layer **62** may differ from the example implementation described herein based on the type of data from the moisture sensor **74**.

(33) Using the determined volume of respective radial layer **62** and the moisture content of the crop material forming the respective radial layer **62**, in combination with the density of the completed first bale **106**, the controller **90** may then estimate a density of the respective radial layer **62** during formation of the second bale **100**. The controller **90** may use machine learning techniques known to those skilled in the art to estimate the density of the respective radial layer **62**. The density of the first bale **106**, in combination with and/or modified by the moisture content of the respective radial layer **62**, may be used to provide an accurate estimate of the density of the respective radial layer **62** during formation of the second bale **100**. As such, the estimate of the density of the respective radial layer **62** of the second bale **100** may be calculated during formation of the second bale **100**, without having to wait until the second bale **100** is completed.

(34) In one implementation of the disclosure, the controller **90** may use the density of more than one previously completed bale, e.g., the first bale **106** to estimate the density of the respective radial layer **62** of the second bale **100**. For example, the controller **90** may use the density of two or more previously completed bales to estimate the density of the respective radial layer **62** of the second bale **100**. In one example, the controller **90** may use a running average of previously completed bales to estimate the density of the respective radial layer **62** of the second bale **100**. In another example, the controller **90** may use a weighted average of previously completed bales, in which more recently completed bales are weighted higher than less recent completed bales.

(35) It is contemplated that the density of the current bale **100** may change as the radial and/or diametric size **58** of the bale **100** increases during formation. For example, the density of the bale **100** may increase as the radial and/or diametric size **58** of the bale **100** increases. As such, the density of each respective one of the plurality of radial layers **62A**, **62B**, **62C**, **62D**, **62E**, **62F**, may change as the radial and/or diametric size **58** of the bale **100** increases during formation. To account for this, the controller **90** may estimate the density of each of the radial layers **62A**, **62B**, **62C**, **62D**, **62E**, **62F** of the current bale **100** during formation of the current bale **100** based on a respective

diametric size **58** of each of the radial layers **62A, 62B, 62C, 62D, 62E, 62F** of the current bale **100**. For example, the controller may estimate the density of each of the plurality of radial layers **62A, 62B, 62C, 62D, 62E, 62F** of the current bale **100** during formation of the current bale **100** by multiplying the estimated density of each respective one of the plurality of radial layers **62A, 62B, 62C, 62D, 62E, 62F** by a density factor. The density factor may be based on the radial and/or diametric size **58** of that respective radial layer **62A, 62B, 62C, 62D, 62E, 62F**. The density factor may include a number equal to or less than one. For example, a respective radial layer **62B** located near the first or center radial layer **62A** may have a lesser density than a respective radial layer **62F** located near the exterior circumferential surface **104** of the current bale **100**. As such, the respective radial layer **62B** located near the center radial layer **62A** may be associated with and the estimated density thereof multiplied by a density factor of, for example, 0.75 to decrease the estimated density of that respective radial layer **62B**, whereas the respective radial layer **62F** located near the exterior circumferential surface **104** of the current bale **100** may be associated with and the estimated density thereof multiplied by a density factor of, for example, 1.00 to not decrease the estimated density of that respective radial layer **62F**. The adjusted values of the estimated densities of each respective radial layer **62A, 62B, 62C, 62D, 62E, 62F** of the current bale **100**, i.e., the estimated density multiplied by the respective density factor, may then be used to define the yield map **98**.

(36) The controller **90** further determines the field origin location **78** of the respective radial layer **62** during formation of the second bale **100**, with data from the location sensor **76**. As described above, the field origin location **78** of the crop material for each radial layer **62** may include a coordinate or other positional identification of the region of the field **46** from which the crop material used to form that respective radial layer **62** of the bale **100** was gathered. The field origin location **78** may include, but is not limited to, an boundary of a region outlining the area of the field **46** from which the crop material of that respective radial layer **62** of the bale **100** was gathered, a centroid location **82** of a region of standardized size representing the region outlining the area of the field **46** from which the crop material of that respective radial layer **62** of the second bale **100** was gathered, or a beginning location **84** and an ending location **86** of a track **88** having a defined width outlining or representing the area of the field **46** from which the crop material of that respective radial layer **62** of the second bale **100** was gathered. The controller **90** may use data from the location sensor **76** to determine the field origin location **78** of the crop material gathered by the pick-up **42** for the respective radial layer **62** of the second bale **100**.

(37) Referring to FIG. 4, the controller **90** may then generate a yield map **98** of the field **46** based on the estimated density and the field origin location **78** of the respective radial layer **62** during formation of the second bale **100**. Accordingly, the yield map **98** may be updated on real time during formation of the second bale **100**, reflecting the yield data from the second bale **100**, before the second bale **100** is fully completed. The yield map **98** may identify the quantity of crop harvested from various regions of the field **46** by calculating the quantity of the crop in each respective radial layer **62** from the density thereof, and associating that quantity of crop to the region of the field **46** from which it was gathered to form that respective radial layer **62**. As shown in FIG. 4, the respective boundaries **80** for each of the plurality of radial layers **62** are generally shown. The yield map **98** may include indicia **108** for each respective boundary **80** indicating the weight of the crop material gathered from that section of the field **46** for that respective radial layer **62**. As noted above, the process described herein may be performed for each of the plurality of radial layers **62** of the second bale **100**. Because the yield data is based on the density of each individual radial layer **62** of the second bale **100**, the granularity and detail of the yield map **98** is increased when compared to a yield map **98** generated only from a fully completed bale **100**.

(38) Additionally, because the yield map **98** may be updated in real time, the controller **90** may be configured to control a function of the baler system **20** based on the yield map **98** of the field **46**. Because the yield map **98** provides highly detailed data during formation of the bale **100**, instead of

post bale **100** processing, the yield map **98** may be used to predict harvest ahead of the baler system **20**, and thereby control the operation of the baler system **20** in real time. For example, the function of the baler system **20** may include, but is not limited to, path planning for the baling operation, weaving control of the baler system **20** to evenly distribute crop material across a width of the baling chamber **34**, a speed of movement across the field **46**, a secondary crop processing function, a forming belt tension, a bale drop location, etc. In order to control one of the functions described above, the controller **90** may be coupled to and configured to control a steering system of the associated work vehicle, a trailing position of the baler system **20**, a pre-cutter setting, a speed of the associated work vehicle, a desired bale density, an bale accumulator associated with the baler system **20**, etc.

(39) The detailed description and the drawings or figures are supportive and descriptive of the disclosure, but the scope of the disclosure is defined solely by the claims. While some of the best modes and other embodiments for carrying out the claimed teachings have been described in detail, various alternative designs and embodiments exist for practicing the disclosure defined in the appended claims.

Claims

1. A round baler system comprising: a pick-up operable to gather and move crop material while traversing a field; a baler system having a baling chamber configured to receive a flow of material from the pick-up and roll the incoming flow of crop material in a spiral fashion into a bale having a cylindrical shape, wherein the bale includes a plurality of radial layers; a bale size sensor operable to detect data related to a radial size of the bale during formation thereof; a bale weight sensor operable to detect data related to a weight of a completed bale; a location sensor operable to determine a respective field origin location of the crop material gathered by the pick-up for each respective one of the plurality of radial layers; a controller including a processor and a memory having a yield mapping algorithm stored thereon, wherein the processor is operable to execute the yield mapping algorithm to: calculate a density of a completed bale with data received from the bale weight sensor and the bale size sensor; determine a volume of a respective one of the plurality of radial layers during formation of a current bale subsequent to the completed bale with data received from the bale size sensor; and estimate a density of the respective one of the plurality of radial layers during formation of the current bale based on the volume of the respective one of the plurality of radial layers, and the density of the completed bale.
2. The round baler system set forth in claim 1, further comprising a moisture sensor operable to detect data related to a moisture content of the crop material for each respective one of the plurality of radial layers.
3. The round baler system set forth in claim 2, wherein the processor is operable to execute the yield mapping algorithm to determine a moisture content of the crop material of the respective one of the plurality of radial layers during formation of the current bale with data received from the moisture sensor.
4. The round baler system set forth in claim 3, wherein the processor is operable to execute the yield mapping algorithm to estimate the density of the respective one of the plurality of radial layers during formation of the current bale based on the moisture content of the respective one of the plurality of radial layers.
5. The round baler system set forth in claim 1, wherein the processor is operable to execute the yield mapping algorithm to determine a field origin location of the respective one of the plurality of radial layers during formation of the current bale.
6. The round baler system set forth in claim 5, wherein the processor is operable to execute the yield mapping algorithm to generate a yield map of the field based on the estimated density and the field origin location of the respective one of the plurality of radial layers during formation of the

current bale.

7. The round baler system set forth in claim 6, wherein the processor is operable to execute the yield mapping algorithm to control a function of the baler system based on the yield map of the field.

8. The round baler system set forth in claim 1, wherein the processor is operable to execute the yield mapping algorithm to estimate the density of the respective one of the plurality of radial layers during formation of the current bale based on the density of at least two previously completed bales.

9. The round baler system set forth in claim 1, wherein the processor is operable to execute the yield mapping algorithm to estimate a density of each of the plurality of radial layers of the current bale during formation of the current bale based on the respective volume of each of the plurality of radial layers of the current bale and the density of the completed bale.

10. The round baler system set forth in claim 9, wherein the processor is operable to execute the yield mapping algorithm to estimate the density of each of the plurality of radial layers of the current bale during formation of the current bale based on a respective diametric size of each of the plurality of radial layers of the current bale.

11. The round baler system set forth in claim 10, wherein the processor is operable to execute the yield mapping algorithm to estimate the density of each of the plurality of radial layers of the current bale during formation of the current bale by multiplying the estimated density of each respective one of the plurality of radial layers by a density factor based on the diametric size of that respective radial layer, wherein the density factor includes a number equal to or less than one.

12. A round baler system comprising: a pick-up operable to gather and move crop material while traversing a field; a baler system having a baling chamber configured to receive a flow of material from the pick-up and roll the incoming flow of crop material in a spiral fashion into a bale having a cylindrical shape, wherein the bale includes a plurality of radial layers; a bale size sensor operable to detect data related to a radial size of the bale during formation thereof; a moisture sensor operable to detect data related to a moisture content of the crop material for each respective one of the plurality of radial layers; a bale weight sensor operable to detect data related to a weight of a completed bale; a location sensor operable to determine a respective field origin location of the crop material gathered by the pick-up for each respective one of the plurality of radial layers; a controller including a processor and a memory having a yield mapping algorithm stored thereon, wherein the processor is operable to execute the yield mapping algorithm to: calculate a density of a completed bale with data received from the bale weight sensor and the bale size sensor; determine a volume of a respective one of the plurality of radial layers during formation of a current bale subsequent to the completed bale with data received from the bale size sensor; determine a moisture content of the crop material of the respective one of the plurality of radial layers during formation of the current bale with data received from the moisture sensor; estimate a density of the respective one of the plurality of radial layers during formation of the current bale based on the volume and the moisture content of the respective one of the plurality of radial layers, and the density of the completed bale; determine a field origin location of the respective one of the plurality of radial layers during formation of the current bale; and generate a yield map of the field based on the estimated density and the field origin location of the respective one of the plurality of radial layers during formation of the current bale.

13. The round baler system set forth in claim 12, wherein the processor is operable to execute the yield mapping algorithm to control a function of the baler system based on the yield map of the field.

14. The round baler system set forth in claim 12, wherein the processor is operable to execute the yield mapping algorithm to estimate the density of the respective one of the plurality of radial layers during formation of the current bale based on the density of at least two previously completed bales.

15. The round baler system set forth in claim 12, wherein the processor is operable to execute the yield mapping algorithm to estimate a density of each of the plurality of radial layers of the current bale during formation of the current bale based on the respective volume of each of the plurality of radial layers of the current bale, the respective moisture content of each of the plurality of radial layers of the current bale, and the density of the completed bale.
16. The round baler system set forth in claim 15, wherein the processor is operable to execute the yield mapping algorithm to estimate the density of each of the plurality of radial layers of the current bale during formation of the current bale by multiplying the estimated density of each respective one of the plurality of radial layers by a density factor based on the diametric size of that respective radial layer, wherein the density factor includes a number equal to or less than one.
17. A method of generating a yield map of crop material formed into a round bale, the method comprising: completely forming a first bale in a baling chamber of a baler system; calculating a density of the first bale; forming a second bale in the baling chamber of the baler system, wherein the second bale includes a plurality of radial layers; determining a volume of a respective one of the plurality of radial layers of the second bale during formation of the second bale subsequent to the completed first bale with data received from a bale size sensor; determining a moisture content of the crop material of the respective one of the plurality of radial layers of the second bale during formation of the second bale with data received from a moisture sensor; estimating a density of the respective one of the plurality of radial layers of the second bale during formation of the second bale based on the volume and the moisture content of the respective one of the plurality of radial layers of the second bale, and the density of the completed first bale; determining a field origin location of the respective one of the plurality of radial layers of the second bale during formation of the second bale; and generating the yield map of the field based on the estimated density and the field origin location of the respective one of the plurality of radial layers of the second bale during formation of the second bale.
18. The method set forth in claim 17, further comprising controlling a function of the baler system based on the yield map of the field.
19. The method set forth in claim 17, further comprising multiplying the estimated density of each respective one of the plurality of radial layers by a density factor based on the diametric size of that respective radial layer, wherein the density factor includes a number equal to or less than one.
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